

Road Map for Implementation of NEP-2020 in HEIs



National Institute of Technology Puducherry
Karaikal - 609609



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PREFACE

The National Education Policy 2020 is envisioned for the paradigm shift in the entire education system of India with the learning process starting from pre-primary education to higher education and continuing the entire life to achieve the ultimate goal of one's life that is liberation. At National Institute of Technology, Puducherry, we wholeheartedly worked out to understand and explore the possibilities for a systematic implementation of NEP-2020 for the benefit of our stakeholders, and to our society.

To propose a roadmap for effective implementation encompassing its full spirit and purpose, the NEP-Cell of the Institute has worked tirelessly under the guidance of the Institute leadership, and along with the faculties from all the departments, and has prepared the comprehensive roadmap to implement the policy in Higher Education Institutions.

After having conducted various invited lectures from eminent educationists and policy-makers, we at NIT Puducherry have come up with a comprehensive note about the "Road map for Implementation of NEP-2020 at HEIs" especially at Technical Institutes of national importance.

It was realized that the benefits of the policy will be imparted to the stakeholder only if it is implemented in a phase wise manner. Such a phase wise implementation and action plan is encapsulated in this document. Equal focus has been given on the core values of the policy namely, Multi-Entry-Multi-Exit system, Choice-Based Credit System, multidisciplinary education, ethos of Indian Culture and glory etc.

With this roadmap, we are looking forward for a phase wise implementation of the NEP-2020 which would emerge as a guiding light for other technical institutes of national importance.

Place: Karaikal
Date: 17-12-2021

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ACKNOWLEDGEMENTS

It gives me immense pleasure to be a part of the preparation of the Road Map for the implementation of NEP-2020 at Higher Education Institute under the able guidance of the Prof. K. Sanakaranarayananasamy, Director NIT Puducherry, one who admires the NEP-2020 and is enthusiastic for implementation from the academic year 2022-23 at our institute.

We thank the Ministry of Education, Government of India for providing continuous support and encouragement to conduct several webinars and explore the different dimensions of NEP-2020, and share the same among the faculty members from institutes of higher education across the country.

This road map document is an outcome of a 5-day national workshop and several brain storming sessions conducted by the NEP cell of NIT Puducherry. The workshop was inaugurated by Prof. Anil D. Sahasrabudhe, Chairman, AICTE, New Delhi. The workshop had insightful lectures, ideas that were helpful in drafting the roadmap document for the implementation. At this outset, I express my sincere gratitude to all the resource persons, Prof. K. Ramachandran, NIEPA, New Delhi; Prof. Sanjiv Kumar, CU Haryana; Prof. S. K. Bhattacharyya, Founding VC of SSN University Chennai; Prof. S. D. Sudarsan, Executive Director CDAC- Bangalore; Prof. Mini Shaji Thomas, Director NIT Tiruchirapalli; and Prof. N. V. Ramana Rao, Director NIT Warangal.

With this, I acknowledge the experience and Ideas shared by the eminent professors, their vision for implementation, all the efforts made by NEP cell, and the NIT Puducherry fraternity to prepare this roadmap document.

Place: Karaikal
Date: 17-12-2021

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Abbreviations

A:	Additional Major
ABC:	Academic Bank of Credit
AEI:	Access, Equity & Inclusion
AMA:	Advance Major
AP:	Advance Project
CBCS:	Choice Based Credit System
CW:	Course Work
DigiLocker:	Digital Locker
DIY:	Do It Yourself
E:	Elective
EAC:	Equivalence and Admissions Committee
FAP:	Flexible Academic Program
ICT:	Information Communication Technology
IQAC:	Internal Quality Assurance Cell
MA:	Major
MDHE:	Multi-Disciplinary and Holistic Education
ME:	Multi Entry
MERU:	Multidisciplinary Education and Research University
MX:	Multi Exit
MXME:	Multi Exit Multi Entry
NAD:	National Academic Depository
NISP:	National Innovation Startup Policy
OBE:	Outcome Based Education
ODL:	Open and Distance Learning
P:	Project
SAT:	Scholastic Assessment Test
SDC:	Skill Development Centers
SEDGs:	Socio-Economically Disadvantaged Groups
SOP:	Standard Operating Procedure
SSC:	Student Support Centers
SSP:	Student Support Programs
STEM:	Science, Technology Engineering and Mathematics/Management
TLC:	Teaching Learning Centre

Introduction

Technical education in India is offered under different autonomous bodies, universities, affiliated institutes and institutes of national importance such as IITs, NITs, IIITs, deemed to be University, polytechnic colleges, ITIs, and other skill development centers established under state and central government schemes. The preamble of NEP brings out the unification of various governing bodies, research funding agencies at the same time autonomous status to the various higher education institutes so that the standards of the education can be uplifted and made accessible to the most deprived class of society irrespective of caste, cadre, region, religion, and gender, etc. Therefore, this document is aimed at understanding the NEP-2020 with the perspective of higher education in general and technical education in particular.

NIT Puducherry is thriving for implementation of the policy from the academic year 2022-23. In this regard, a workshop was conducted on “Road map for implementation of NEP-2020 in HEIs”. The workshop had focused on lectures from eminent academicians, administrators who have been involved in drafting the policy and started the implementation of the policy in their organizations. The extensive question and answer sessions, brainstorming sessions have inspired us to prepare this document that applies to higher education institutions.

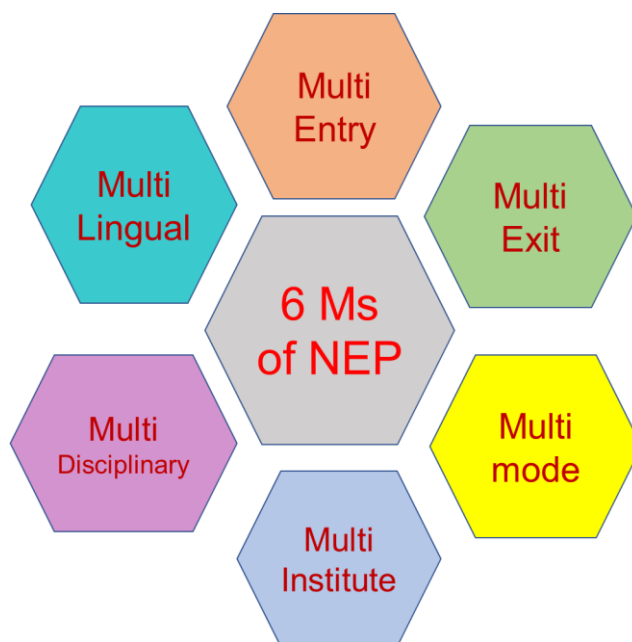
The major outcome from the implementation of NEP-2020 in higher education institutions is to generate a backend technical support system for the industry 4.0 revolution and to establish world-class opportunities for international students aspiring to study in India. This document provides guidelines for implementation of NEP-2020 in higher education institutes. The proposed guidelines can be adopted by higher education institutions for an effective and efficient implementation of the policy to contribute for transforming India into a ‘Vishva Guru’.

Orientation towards NEP-2020

Aspirations of students to take up technical education and engineering as a career are vast and multifarious. At present students are getting admitted to engineering institutes either by passion or else as an option or for societal prestige. However, a majority of desirous students could not take up their choices due to various reasons such as, lack of awareness on the scope of engineering discipline, lack of privileges (financial, societal, etc.), and lack of clarity on employability due to tremendous Industry 4.0 revolution. A few of the features mentioned in NEP-2020, which all unique in the Indian context are already being practiced in conventional higher education system. The students can experience the holistic features of learning through following the proper orientation mechanisms:

- Organizational services for holistic learning: In this regard, academic programs need to be strengthened with tailor-made pedagogical practices aligned to the mode of learning, leading towards quality education. In this regard, scope shall be given for assessment through creative evaluation techniques at different difficulty levels such that a student can take them as a part of learning.
- Enhanced academic support such as university wellness through alumni, Industrial collaborations, student services, etc. Such facilitation may help the students to access and transit from various educational levels. Synchronize their curricular and co-curricular activities at various levels to enhance their skill, which is an essential task in contemporary education.
- Bridging the Socio-economic barriers: “Education for everyone” can be realized through enhancing the existing socio-economical supports like Scholarship portals, Women empowerment, establishing friendly infrastructure for the physically challenged students. Such facilities need to be offered at all the stages of learning process, with transparency to realize their deserving profile and self-confidence.
- Integration of Technology and Heritage: The technology-enabled pedagogic approach for learning at different institutes is a vital component. To

strengthen the students' aptitude, it is essential to recognize the value of vast cultural heritage of India. Further, the learning outcome can be enhanced through accessing the multi-lingual learning resources through technology-enabled platforms. These aspects make both the institute and the students to inculcate an approach for various locally challenging issues that need technological intervention. In summary, a student will get adapted to a curriculum that encourages the development of research and thus will develop a personality that values science, technology, innovation, society and heritage. To achieve these objectives, NEP-2020 brings out the optimal solution for such issues, which can be presented in terms of 6 Ms.



For effective implementation of NEP-2020, the curriculum needs to be redesigned and aligned with requirements of the industry. Therefore, the students need to be re-oriented towards outcome-based learning using different learning modules. The following learning modules will help them to attain holistic development, which is the “the primary objective of NEP-2020”.

Modes of learning

- **Field based:**

In field-based learning, learner is asked to gain hands-on application of the course content through interaction with the environment rather than traditional classroom lectures or pre-designed laboratory exercise.

- **Self-study:**

In this mode of study, a student will be given the outline of topic to contemplate by themselves to assimilate the knowledge.

- **Flipped classroom:**

The concept will be introduced prior to the class, to increase the student engagement with subject at home and explored at classroom through live problem-solving technique.

- **Online learning:**

Online mode of learning is completely technology enabled; delivery, interaction and assessment will be conducted virtually. Online classes can be attended at the convenience of the learner through portals such as SWAYAM, NPTEL, MOOC etc.

- **Community engagement:**

Learning by engaging self with the society will lead to learning for the life time. Such learning methodology will be thought provoking and bring out the emotional and intellectual strength in the learner, leading to innovative and creative solution to the prevailing issues.

- **Internship and Research:**

This is a unique opportunity for the learner for applying his/her skills for industrial or real-life problems. This mode of learning brings field experience and gives confidence in career building.

- **Assessment and evaluation**

- **Allocation of credits:**

Choice Based Credit system (CBCS) shall be introduced for awarding the credits towards the completion of course. The students will earn credits for the registered courses. Once the student acquires sufficient credit points he/she will be awarded with the relevant certificate/degree.

- **Benchmarking the assessment and progress in learning:**

The learning outcome is realized through assessment process. This can be formative and/or summative in nature. Formative assessment will be held during the instruction period. Outcomes of the formative assessment will be helpful in strategizing the instructional methods. On the other hand, the summative assessments may be held at the end of a course to estimate the overall learning.

- **Scope for fast and slow learner:**

NEP-2020 is a learning centric policy which gives importance to Outcome Based Education (OBE). The hard separation in the results such as 'Pass' and 'Fail' is replaced by degree of skill attained by the student. A slow learner can select the courses as per his/her capability and attain the learning. On the other hand, a fast learner can select his/her courses and meet the requirement of academic credits required for award of the certificate/degree.

Execution

➤ **Mechanism for Multiple Exits (MX):**

- Mapping of courses at intra-institute levels, and also at inter- institute/ inter university levels will be done. This will enable the students to continue their studies hassle free through different modes.
- Identifying a uniform certification process for the programs at exit levels, and mapping the equivalence of degrees. With this provision, the diverse knowledge acquired by the student will get unified towards career prospective and employment opportunities.
- For the programs where the immediate exit option is not feasible, in such cases the flexibility in exit can be introduced by the following mechanisms:
 - ❖ Provision for registering for a course through alternative platforms.
 - ❖ Initiating several new courses at primarily enrolled parent institute, so that the student can earn maximum number of credits at a single institute, to avoid the conflict of interest in certificate/degree awarding institute.

➤ **Mechanism for Multiple Entry (ME)**

- Students enrolled for the first time to the institute need to be oriented towards scope and all provisions of multiple exits and subsequent entries. Such a feature

can be enabled by creating a nodal agency to streamline credit transfer across the nation and to make assessment of acquired skill sets hassle free.

- The aspiring students have to full fill certain pre-requisite knowledge and skill sets to take up a particular course of choice. Therefore, the student has to undergo bridge courses.
- Bridge courses are mandatory for multiple entries to an institute.
- Mapping the academic credits earned by the students to an equivalent program of the primarily enrolled parent institute for subsequent award of the certificate/degree.

Flexible Academic Program (FAP)

NEP-2020 provides scope for flexibility in earning academic credits through different modes of learning to enable MXMEs (Multiple Exit- Multiple Entry) culture. However, technical institute have to assign minimum number of credits to be earned by the student for award of a certificate/degree. In this process, the students have to undergo following learning process to realize the MXMEs:

- Finishing school/ Bridge courses to orient the student aspiring for admission under MXME mode.
- In-house opportunities primarily at enrolled institute for start-up or for employability while exit at any level.
- NEP-2020 enables plenty of opportunity for every learner from undergraduate program to Ph.D. program. Therefore, the curriculum will be tailored in accordance with objectives of the program offering department in the particular discipline, and vision-mission of the institute. Thus, during MXMEs the student have to undergo the finishing school/ bridge courses to become eligible for higher studies, as well as enhance their skills to meet the employability criteria, and also become a valuable resource to the society.

The execution aspects of the 6 Ms mentioned in the previous section lies in the successful implementation of the Flexible Academic Program (FAP). The entire process of higher education is the interconnection of learning and the development of competencies, academic mobility at various stages of exits and entries, and integrating the diverse educational options to converge at the level of Masters and Doctorate degrees. The key aspects of FAP with reference to the

technical education from admission to an undergraduate program up to Ph.D. is as follows:

Semesters	1	2	3	4	5	6	7	8	9-12		13-16	
Minimum Credits	20	44	60	84	100	125	140	160/ 172	200/250		350	
Multiple Entry points	ENTRY		ENTRY		ENTRY		ENTRY	ENTRY		ENTRY	ENTRY	
Student Support Programs (SSP)	Intensive Orientation	Finishing School	Bridge course	Finishing School	Bridge course	Finishing School	Bridge course	Finishing School	Finishing School	Bridge course Finishing School	Bridge course Finishing School	Pedagogy training
Degree/ Certification		Certificate		Diploma		B.Voc./ B.Sc.(Engg.)		B.Tech.	B.Tech. Honors	M.Voc./ M.Sc.(Engg,)	M.Tech.	Ph.D.
Multiple Exit points		EXIT		EXIT		EXIT		EXIT	EXIT	EXIT	EXIT	EXIT

- **Semester:** This is a traditional half-year term of about six months, which may take 18 to 20 weeks of academic instructions.
- **Credits:** Academic recognition to quantify the efforts and learning process that student has experienced for a given course offered in a program.

The academic credits gained by the students will be managed through National Academic Depository (NAD). Thus, the digitally created academic credits stored at NAD are accessed through the DigiLocker, which ensures easy access and retrieval of academic records ensuring authenticity and safe storage.

- **Certificate/Degree:** Qualification earned by a student upon achieving academic and professional competency, and thus earning threshold of academic credits. For the academic semesters 1 to 10, the credits are allotted for the academic learning and development of proficiency in skills. On the other hand, for the

semesters 11-16, students need to inculcate skills for education and research among themselves. In any circumstances, if the students fail to accumulate the required credit points for the award of degree, they will be mentored by the faculty for the eligible lower certificate or degree based on the available earned credits.

NEP-2020 encourages multiple exits and multiple entries, with the idea of 'light but tight' assessment system. Therefore, exit is allowed at certain semesters where the student can take crucial decision to sustain professional and academic opportunities in the future. Similarly, the entry points indicate direct re-entry for the any students, as well as for the student aspiring for lateral entry. Hence, in the academic life cycle the exit is never considered as a 'Dropout' rather, the learning is considered as life long process.

➤ **Student Support Program (SSP):** A student support program refers to a framework for the stakeholders to facilitate quality environment, thorough follow-ups, and opportunities for employment/employability, life skills, and other supports related to academic performance. The main features of these programs are as follows:

- Networking support to the students for effortless adaption with the university life during the entry stage.
- Strengthening the relationship with interest groups such as cooperation bonds with the admitted Institution and various agents of the industries, institutions with MoU, and civil society, etc.
- Management of graduation indicators, strengthening the technical and soft skills of the students for self-reliance until they become responsible and successful citizens that generate value in society.
- Management and control from termination of the course to “student success”, by ensuring a respectful learning environment.

➤ **Orientation program:**

The institute of higher education has to offer a mandatory intensive orientation program 'Deeksharambh' to the students admitted for an academic program.

This course is to be offered during the very first semester of the undergraduate program. The course will be unique in nature to address the preparedness of a student to take up different aspects of NEP such as 6Ms, assessment, flexible academic program, learning outcome and exit processes. This will lead to a foundation for planning their education process. In summary, this program aims to provide a smooth transition from school education to higher education.

➤ **Finishing School:**

Any student intending to exit from the admitted academic program will have to undergo a mandatory course called 'Finishing School'. This is an Institutional support program to analyze and identify their strengths, weaknesses according to the available opportunities for them based on their academic performance, skills, psychological risks, principles and values. Accordingly, requisite training will be provided to enable them for professional career or continue their education in the discipline of their interest. The duration and content of this course will vary according to the exit stage from certification (40 credits) to M.S. by research (250 credits). An academic certificate or degree is awarded to the student only after completion of this mandatory finishing school course. Thus, the student emerging out from the institute will ensure as a value addition to the society. The student registered for Ph.D. program will undergo a holistic pedagogic training suitable for practicing the profession independently in industry and academia.

➤ **Bridge course:**

Bridge course is offered at each entry-level opted by the student. This program is complementary to the orientation program (offered at the first entry) to orient the aspiring student to the organizational structure, vision- mission of the currently opted department and institute. Special emphasis will be given to the analysis of strengths, weaknesses, and competency of the students to ensure the pre-requisite of the courses. Based on the academic background and performance acquired in the previous certification/degree or professional experience gained between the exit and re-entry, capabilities will be enhanced

through the bridge courses. In case of lack of competencies, the student may be mentored to attain his aspirations.

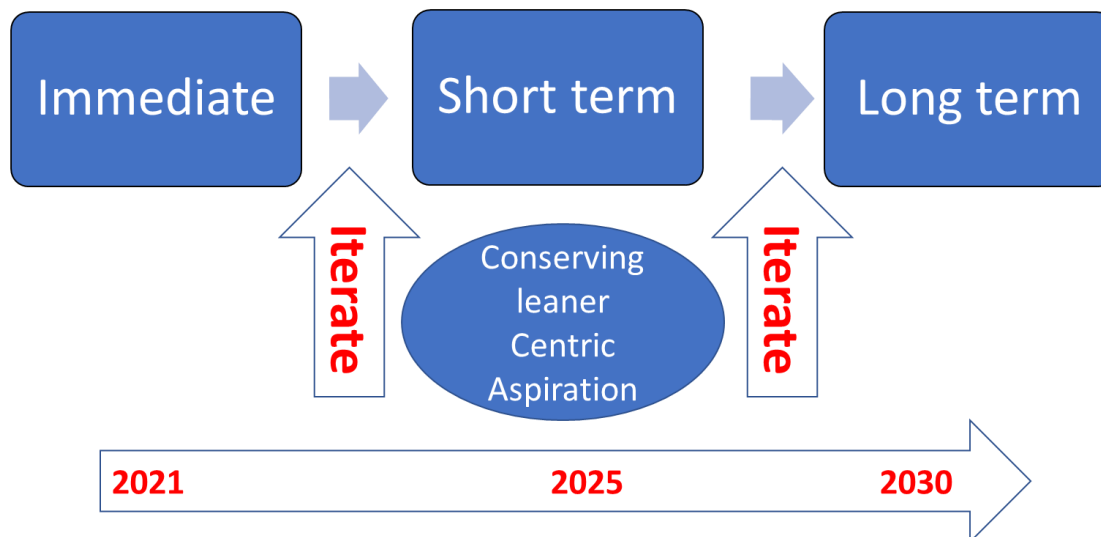
In case of lateral entry, all the requests shall be considered by the Equivalence and Admissions Committee (EAC). The institute may decide to offer the course at a particular entry node through in campus or through Open and Distance Learning (ODL) modes. This decision will be taken based on the nature of course and also based on the strengths, weaknesses and academic competencies achieved by the student, and moreover availability of academic infrastructure to accommodate ongoing batch of students. In case, their admission is confirmed, such students will avail the institutional support and mentoring system until they exit from that academic course.

Action plan

Implementation of the NEP-2020 policy brings transformation of the present school education system 10+2 in to 5+3+3+4 (includes 3 years pre-school). Accordingly, by the end of the upcoming decade, the number of students aspiring for education will have basic technical education as a part of life style irrespective of their nature of higher education. As the policy aspires successful implementation by 2030, reformation of education in HEI has to be implemented simultaneously within a decade, Therefore, different facets of technical education need to be reformed accordingly to meet the requirements of the students.

➤ Immediate action plan

- Frame work for mapping of academic credits among HEIs and the intuitions of National importance through Academic Bank of Credit (ABC).
- Establishment of office for international students to spread awareness and provide support.
- Grant for establishment of infrastructure of existing NITs/ Technical institutes to facilitate them to become Multidisciplinary Education and Research University (MERU).



- Financial grant for introduction of multidisciplinary departments such as Medical and allied disciplines, Fine Arts, Humanities, Management in existing technical institutes of National importance.
- Establishment of regional level state of the art infrastructure for creating online courses.
- Preparation of resource material to offer education in vernacular languages.
- Teaching Learning Centers (TLC) for School teachers and faculty members from government/ private affiliated colleges.
- Registration on academic bank and credits (ABC).
- Setting up of Internal Quality Assurance Cell (IQAC) at each institute with Standard Operating Procedure (SOP), which will ease the accepting of the credits during the multi-institute learning.
- Identifying the institutes/ Industry who offer courses and are ready for sharing credits, signing MOUs in this regard etc.
- Enhancing the competency and capability of institute / faculty to initiate the implementation of NEP-2020 effectively in their institutes and guiding other institutes.
- Drafting general rules and guidelines for enabling the students to take up mandatory internship in industry, and assessment of the training outcomes to grant the relevant grades.

➤ **Short term action plan**

- Development and implementation of experimental learning curriculum.
- Initiating Industry relevant courses.
- Formation of Equivalence and Admissions Committee (EAC).
- Integration of curriculum suitable for multiple exits and multiple entries.
- A mechanism to monitor the balance between the technical education and employability.
- Opportunities for strengthening unexplored domains of the faculty.
- Usage of Information Communication Technology (ICT) Tools in classrooms/ laboratories.
- Mentoring the neighboring institutes/ teachers for STEM (Science, Technology Engineering and Mathematics/Management) program.
- Implementation of a policy for the students to take up short-term/ free-lance opportunities.
- Setting up the regulations in accordance with National Innovation Startup Policy (NISP), enabling the students to take up start-up activities, in lieu of academic credit points.
- Establishment of Self learning and do it yourself (DIY) Laboratories.

➤ **Long term action plan**

- Mechanism for setting up testing proficiency of entry level criteria, similar to SAT that enables the students to take up multiple entries / and multiple Institute entries.
- Prioritizing the courses of contemporary and global relevance in the elective baskets that emerge periodically.
- Visiting faculty from collaborative institute/ Industry. Internationalization of the institutes and setting up of satellite campuses for enabling the faculty to offer courses and enabling the students to avail quick access to the facilities and immediate assistance from the institute.

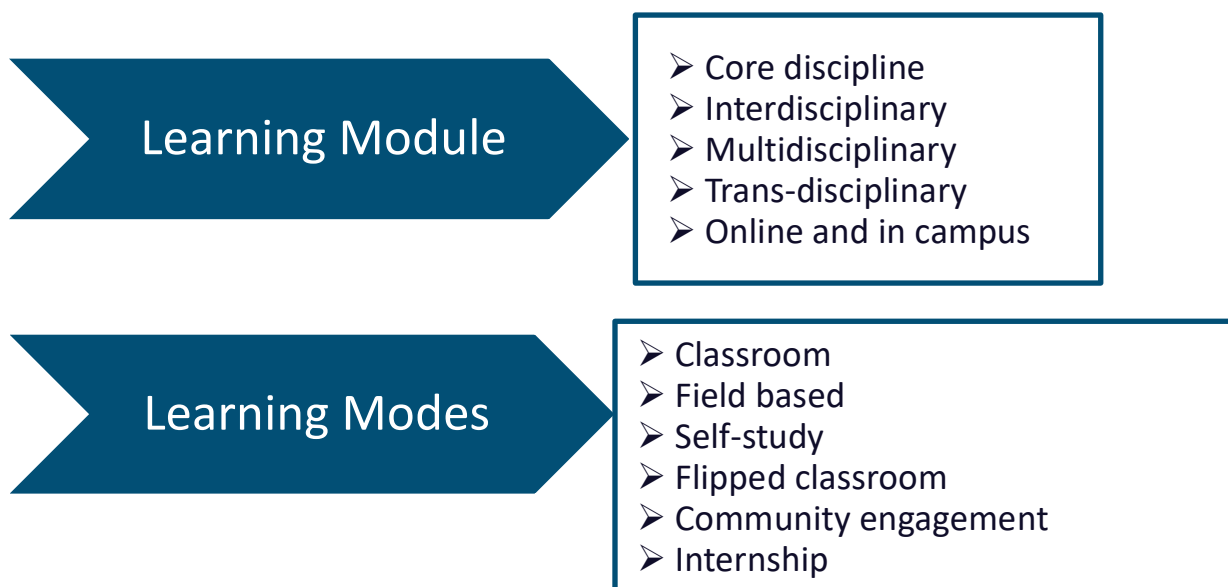
Role of Technical Institutions

Implementation of NEP-2020 in school education will take in a span of 15 years, whereas in the higher education it takes span up to 10 years, including the maximum academic degree of Doctor of Philosophy (Ph.D.). Implementation at the higher education institute is quite challenging as the academic programs are diverse in nature. The key domains to be focused for effective and immediate implementation of NEP-2020 are as follows:

- Industrial and University linkage through MoUs for offering courses by industry experts complementary to the academic learning and for offering bridge courses and finishing schools.
- Pedagogical model and ICTs for academic support, development of Institutional models for the students and faculty.
- Engagement with alumni to avail real time feedback throughout the implementation of the policy.
- Cooperation with non-technical/ vocational institutes.
- By law for institute which institute / University offers the degree.
- Translation, teaching-Learning centers, state of art facilities for creation of learning materials.

➤ Multi-Disciplinary and Holistic Education (MDHE)

The following model may enable the students to avail multi-disciplinary and holistic education



Assessment

- Benchmarking learning progress
- Scope for fast and slow learner
- Allocation of credits
- Mechanism for multi exit and multi entry

➤ Multi-Exit and Multi-Entry (MXME)

NEP promotes multi-mode learning process as explained in earlier sections. The multi-mode learning option enables experienced learning, staying tuned with MXME process. Unlike the traditional delivery of lectures in classrooms, in MXME (Multi Exit and Multi Entry) course content mentioned in the curricula will be taught in online mode, individual-centric contact courses, and few classroom sessions. Thus, the student undergoes several learning processes such that he/she found suitable for multiple exits and undergo some advance courses to match the need of the employer. In this regard, tutorials, learning material should be made available to the student. According to the above FAP, detailed allotment of credits for various entry and exit points is proposed below:

Table 1. Possible combination of Major and Minor with Multi-Exit system

Duration	Degree	Credits required	Description
1 Year	Certificate	44	SMASH + E
2 Years	Diploma	84	SMASH + MA +AP1 + E
3 Years	B.Voc./B.Sc.(Engg.) with Major in __	125	SMASH + MA1 +AP2 + E
3 Years	B.Voc./B.Sc.(Engg.) with Major in __ and Minor in _	125	SMASH + MA1 +AP2 + E (Minor basket (20-26 credits) + Electives (if credit requirement is not satisfied))
4 Years*	B.Tech. with Major in __	160	SMASH + MA1 + AP2 + subjects from any basket to satisfy 61 credits
4 Years*	B.Tech. with Major in __ and Minor in __	160	SMASH + MA2 + AP2 + one minor basket of 20-26 credits and remaining credits from electives
4 Years*	B.Tech. with Major in __ and Minor in __ and __	160	SMASH + MA1 + AP2 + two minors basket of 20-26 credits and remaining credits from electives

4 Years*	B.Tech. with Major in __ and Major in __	160	SMASH + MA1 + AP2 + MA2 + remaining credits from electives
5 Years	M.Voc./M.Sc.(Engg.) in __	200	Carry forwarded 160 credits + 22 credits for advanced courses in the chosen specialization + 18 project/internship
6 Years	M.Tech. in __	240	Carry forwarded 160 credits + 22 credits for advanced courses in the chosen specialization + 18 project/internship + 40 credits for Research work
8 Years	Doctor of Philosophy	350	Carry forwarded 200 or 240 credits + 20 credits for pedagogy training + (130 or 90) credits for Research work

*In Case of Honors candidate has to do additional 12 credits from the respective chosen major(s).

- Candidate can opt for minor basket of 20-26 credits in place of E.
- Major 1 belongs to candidate's parent department.
- Major 2 belong to the department other than candidate's parent department.

Assessment is an integral part of teaching –learning process. Assessment can be formative and summative. The former method is adopted throughout instruction process, for improving learning outcome and to strategize the delivery. On the other hand, summative assessment is implemented at the end of the course to consolidate the learning outcome of the semester. Traditionally, the formative assessment has less weightage compared to summative assessment. As NEP-2020 is learner-centric and enable slow and fast learner the formative assessment has to be given importance. Thus, summative assessment becomes deciding factor to check whether student attained program outcomes required for employability.

Table 2. Course structure and credit requirements

Description	Details	Total Credits	Comments	Exit Degree						
Mathematics + Science + Humanities + Arts	SMASH	35	Science+ Mathematics + Arts + Social Science + Humanities (SMASH) Basket	Certificate in ____	Eligible for Diploma in ____	Eligible for B.Sc. in Engineering Degree with Major in ____ and/or minor in ____	Eligible for B.Tech. Degree (4Years) with Major in ____ and ____ or Major in ____ and minor in ____ and ____	Eligible for B.Tech. With Honours Degree (4 Years) with Major in ____ and / or Minor in ____	Eligible for Postgraduate Degree M.Tech. with specialisation	Eligible of Doctor of Philosophy Ph.D. Degree
Major	MA	40	Major for 2-year Diploma							
Total (C)		44	Maths+ Science+ Skill based learning							
Project and Internship	AP1	6	If opting for 2-year course							
Electives	E	3	If opting for 2-year course							
Total (D)	SMASH + MA1 +AP1 + E	84	Mandatory Credits for 2 Years Course							
Major	MA1	MA+12 = 52	Major Basket-1							
Project and Internship	AP2	12	In case of Two major the project and internship can be done in any of the one major							
Electives	E	26	One minor or electives							
Total (B1)	SMASH + MA1 + AP2 + E	125	Mandatory Credits for 3 Years Course							
Additional Major / Elective	MA2	61	Major or 2 minors and/or Electives							
Total (B2)	SMASH + MA1 + MA2+ AP2 +E	160	Mandatory credits for 4-year course							
Honours*	Theory	12	Additional subjects							
Total (B3)	Total	172								
Advance Major	Courses AMA1	22	Advanced level theory courses for specialization							
Advance Project	Project AP3	18	Project with Research Component							
Total (M)	B2+ AMA1 + AP3	200	Mandatory of PG Degree							
Mandatory course work	CW1 + Pedagogy	20								
Research	Research Project RP	90 / 130	90 for admission after 6-year course (Refer Table 1) 130 90 for admission after 5-year course							
Total	M + CW1 + RP	350								

*In Case of Honors candidate has to do additional 12 credits from the respective chosen major(s).

➤ **Socio-Economically Disadvantaged Groups (SEDGs)**

Education for everyone can be realized through **Access, Equity & Inclusion (AEI)**. This can be achieved by providing infra-structure facilities to the physically disabled student, empowering the women and socially deprived through special sensitization and grievance redressal programs, setting up of Student Support Centres (SSC) for counselling, physical and psychosocial wellbeing. Accordingly, institutions need to focus on following key aspects where learning can take place without the barriers like caste, cadre, gender, religion and regions etc.

Focused Mentoring

The student will be mentored by the Teacher, who takes role of the 'GURU' as per the ancient concept of Indian philosophy. Here, teachers will not only guide the students at each exits and entries, but also be able to assess the proficiency of the student.

Bridging ancient and modern knowledge

The curriculum has to be hybridized with current advancement and ancient knowledge. This type of curriculum will enable the student to know and incorporate ancient knowledge into his/her innovative ideas and research.

Professional and life skills

For successful implementation of NEP both teacher and the students have to enhance the teaching skills for online platforms, lesson plans and script for video recorded classes or e-class rooms. Also, the teacher has to learn to disseminate the skill sets for balancing personal and professional life.

Vocational and general education

Vocational education gives scope for immediate employment, however to enhance the opportunities in future career, it is required to adapt the technological and structural change. It can be addressed by providing theoretical knowledge to the student who seeks for lateral entry. Similarly, the student who has acquired conventional education and opt for exit should be provided with appropriate skills at par with vocational courses. In this regard technical institutions may establish Skill Development Centers (SDC).

Traditional and innovative pedagogy

The faculty has to inculcate both traditional practices of teaching as well as modern e-classes and e-resources for assessments. Teacher may need training for setting up of personalized e-classroom arrangements.

Curricular, co-curricular and extra-curricular activities

Major goal of the NEP-2020 is holistic development of personality while learning. Therefore, the combination of curricular, co-curricular and extra-curricular activities needs to be inculcated in the available program. Competency in sports/ fine arts/ management skills needs to be benchmarked, with the cooperation of non- technical/management institutes.

Institutional Development Plan

Infra-structure facilities available institutions make the students to opt for multi-institutional learning process. The facilities available at each institute need to be standardized to make the learning multi-mode and multi-institutional. In view of these following aspects has to be addressed by the HEIs

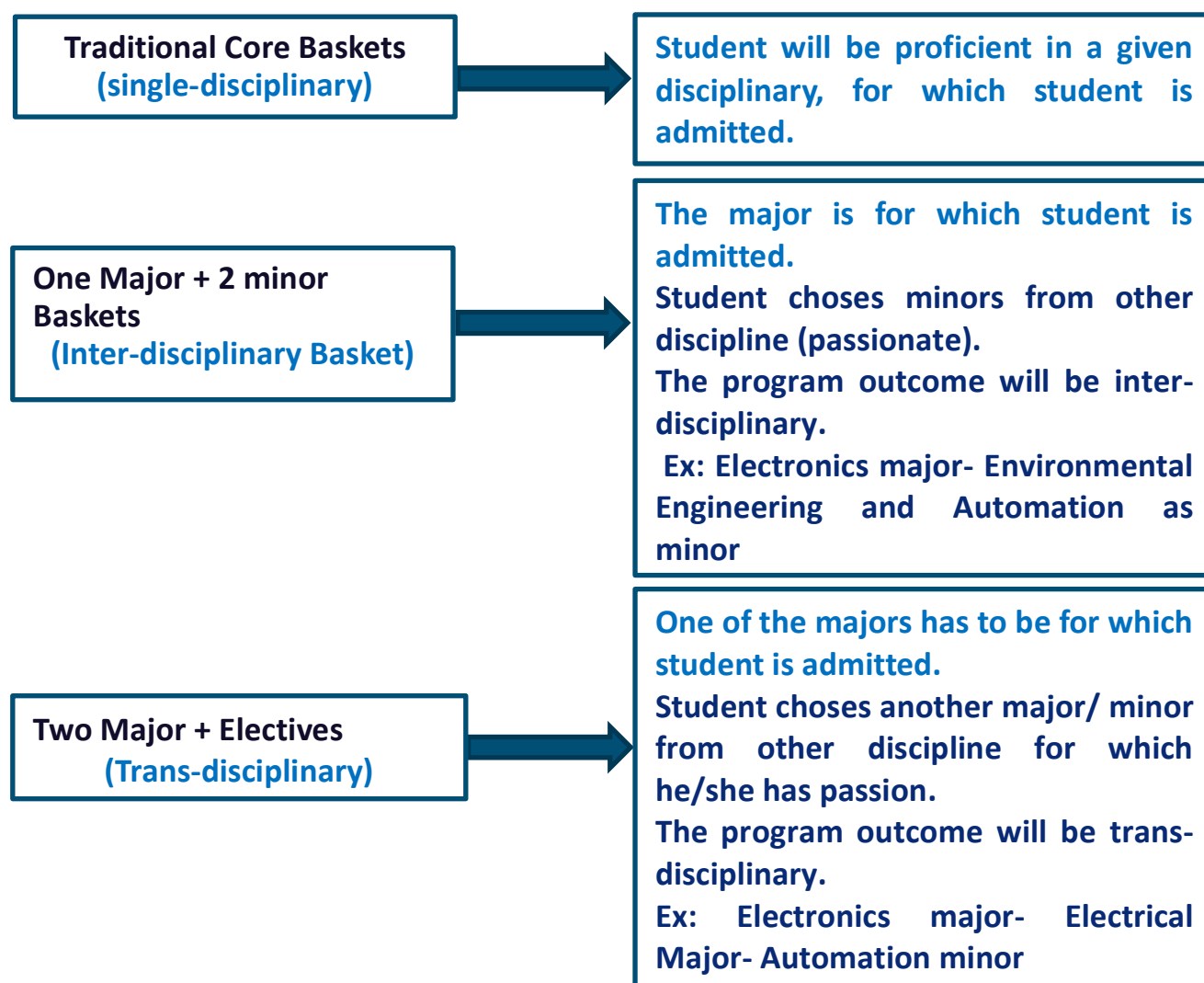
- Establishment of infrastructure of existing NITs/ Technical institutes to facilitate to become Multidisciplinary Education and Research University (MERU).
- Introduction of multidisciplinary departments such as Medical and allied disciplines, Fine Arts, Humanities, Management in existing technical institutes of National importance.
- Establishment of regional level state of the art of infrastructure for creating online courses.
- Establishment of virtual laboratories, open source software's for analysis of data.
- Establishment of Institute-Industry linkage for research, internships and employments.
- Establishment of Institute and school education linkage through Teaching Learning Centers (TLC), to train the teachers to disseminate the scope of NEP in higher education, innovative pedagogy system, carryout research and design school education curricula, and mode of teaching in accordance with emerging international trends.

- Setting up of unique mechanism to address to need of industries and redesigning the curricular structure as per the evolving trends in technology. This will encourage the learners to take an option of multiple exits and entries.

A model for offering the courses

➤ Learning Modules:

Learning modules can be designed by providing wide option for choosing the courses for a given program for which a student has been enrolled in a university or institute. The scheme provided below is 'light but tight' as it does not deviate from the program outcomes that a student has chosen with opportunity for undergoing the course that he/she is passionate about. A student enrolled at a given institute can make the following options.



The above mechanism balances the eco-system of professionals in the nation. Also serves as remedy for present system of admission through competitive examinations as per the seat matrices approved by various governing/ autonomous administration. However, in addition to the above choice of courses, the student has to undergo following courses for overall proficiency with human values.

- Professional Ethics code- constitutional values.
- Intellectual Property Policy awareness.
- Skills for conservation- Employment generation.
- Internship /Socially relevant project/ start up activity.
- Inter-personal relationships.
- Aptitude for socio-economic responsibilities.

The student has to undergo certain mandatory courses and proficiency levels, prior to award of a degree/ certificate at any exit point. This aspect seems to be 'tight' but in line with the vision and mission of NEP-2020. The depth of course content has to be tailor made in accordance with the different levels of learning outcomes mentioned in Bloom's taxonomy (Outcome Based Education) for each qualifying exit programs. Similarly, if the student is opting for Ph.D. he/she has to acquire the fundamentals of teaching pedagogy.

➤ An Illustration:

As per the Table 2, the minimum credits requirement for the award of 4-year B.Tech. degree is 160. The following case is applicable for 4-year B.Tech. With one Major and one Minor. If a student is admitted to the department of Civil Engineering (CE), the candidate's 1st Major will be from his parent department i.e., CE. The candidate has to take a major of 52 credits from his parent department. In addition to this the student will acquire 30 credits from SMASH courses and 12 credits of project and internship. Thus, the candidate earns 94 credits and requires additional 66 credits. Now, the student has the freedom to choose the courses as a combination of major and minors or combinations of minors/ electives offered by the other departments. If candidate opts for only one minor (26 credits), in that case the candidate accomplishes $94+26=120$ credits and needs another 30 credit.

The remaining 30 credit requirement can be satisfied by taking courses from any electives (global or departmental).

Conclusion

Outcomes from the implementation of NEP 2020 in school education can be visualized only after 12 years. On the other hand, the industry 4.0 revolution requires immediate technological support from the young generation. In this context, implementation of the policy in institutes of Higher education is very essential. The other presumed outcomes of the policy are excellence in academics through sports, and management and thus to make our motherland India as a superpower in the academic sector. In a continuously evolving system, changes are inevitable. Therefore, one has to adopt the new avenues and adapt the outcomes for sustainability in higher education system. Thus, this document comprising of a road map will serve as a handbook for many institutes for immediate and effective implementation of the policy from the academic year 2022-23.

WORDS OF GRATITUDE

NIT Puducherry has taken a unique initiative to understand the key aspects of NEP-2020 pertaining to Higher Education. As a first step towards this, NEP cell of NIT Puducherry has organized a 5 days' workshop to understand the gist of the policy through presentations by eminent educationists.

The workshop was conducted in hybrid mode, in person for the faculty members of NIT Puducherry and online mode for outside participants. The workshop has a strategic theme and received huge response with more than 100 participants from various institutions of higher education including senior faculty members from other NITs, and universities.

Every session of the workshop had extensive question and answer sessions with the experts, discussions which have brought more clarity to lay down this document. At the outset, we thank all the resource persons from our bottom of heart for readily accepting our invitation, patiently clarifying all the queries raised by the participants. We are very much thankful to the faculty members from other NITs and institutes of higher education for their active participation.

We extend our sincere thanks to the faculty and staff members of NIT Puducherry for their participation in several hours of brain storming sessions in designing the different modules of this document applicable for HEIs.

We acknowledge and appreciate the efforts taken by Mr. Shivaraj Singh, III year B.Tech- Computer Science Engineering department for designing beautiful back cover with a theme of NEP-2020. In this context, we also thank the members of the Design club, NIT Puducherry for their valuable inputs.

We sincerely acknowledge the support from our Director Prof. K. Sankaranarayansamy and Prof. G. Aghila, Registrar (i/c) for their support and guidelines for conducting this workshop, and providing their valuable suggestions in drafting this document on "Road Map for Implementation of NEP-2020 in Higher Educational Institutes".

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NEP 2020

Road map for HEIs

Multi-Entry

Multi-Mode

Multi-Exit

Multi-Institute